

Results

With a mean follow up of 13.51 months (range 3.83-18.93) the local control was 100%. There were no G3-4 intraoperative or postoperative complications. Acute toxicities were wound infection in 3 patients and poor healing with wound dehiscence in 5 patients. No late toxicities were seen.

Conclusion

IOHDR brachytherapy using the Freiburg flap is a feasible, safe and well-tolerated component of multimodally therapy for different types of tumours. Allows dose escalation in a well-defined intraoperative surgical bed, separating or protecting the organs at risk, reducing EBRT dose and the side effects. Although the results are encouraging we need further follow up.

EP-2254 Rapid learning in a distributed ecosystem: modeling maculopathy occurrence after eye brachytherapy

A. Damiani¹, C. Masciocchi¹, N. Dinapoli¹, G. Chiloio¹, L. Boldrini¹, J. Lenkiewicz¹, M.A. Gambacorta¹, L. Tagliaferri¹, R. Autorino¹, M.M. Pagliara², M.A. Blasi², R. Gatta¹, R. Negro³, J. Van Soest⁴, A. Dekker⁴, V. Valentini¹

¹Università Cattolica del Sacro Cuore - Fondazione Policlinico "Agostino Gemelli", Radiation Oncology Department, Rome, Italy

²Università Cattolica del Sacro Cuore - Fondazione Policlinico "Agostino Gemelli", Ophthalmology Department, Rome, Italy

³KBMS srl, Radiation Oncology, Rome, Italy

⁴Maastricht University Medical Center, Radiation Oncology MAASTRO-GROW School for Oncology and Development Biology, Maastricht, The Netherlands

Purpose or Objective

Distributed Learning (DL) is a method which trains prediction models (PMs), leaving all patient (pt) data at the institutions that collected them, obtaining the same results as the centralized approach. Our aim is to train a DL PM using real clinical data and a real Rapid Learning Infrastructure (RLI) implementing a "distributed ecosystem" in which preliminary dataset analysis tools, DL run, and automatic PM update, are three steps toward a robust, constantly updated PM in a "Really Rapid Learning" environment.

Material and Methods

To test our Distributed Ecosystem Prototype, we used the Varian Learning Portal (VLP) simulator for deployment of algorithms in 3 sites, and aggregation of site PM results with automated PM production and update, at a central server (DOI:10.1016/j.j.ctro.2016.12.004). Pts from institutional database (DB) affected by choroidal melanoma and treated with Ruthenium-106 plaque from December 2006 to December 2014 were selected. Inclusion criteria were dome-shaped melanoma, distance to the Fovea (DF)>1.5 mm, tumor thickness >2 mm and follow-up>4 months. Prescribed dose was 100 Gy at tumor apex for all. Presence of diabetes, tumor volume and distance to the fovea were chosen as factors, with the occurrence of maculopathy as outcome. The collected DB was then randomly split in 3 parts, to simulate 3 different data originating institutions. PMs were developed to be autonomously executed on each DB in order to learn the DL Cox PM by using the RLI. Beta coefficients and p-values of DL and centralized cox PM were compared. The PM was automatically published on an interactive website, is updated when new patients are added, accessible and executable as a Decision Support Aid (DSA).

Results

197 pts with a median age of 68 (range: 17-92 years) were considered for this analysis. The median follow-up was 51 months. The occurrence of maculopathy was recorded in 21% pts after treatment. The PM was trained

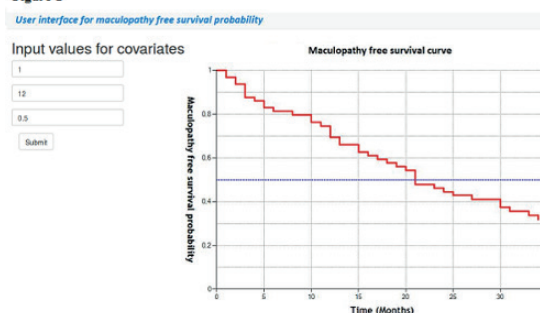
using real RLI. Table 1 shows that the same beta and p-values were obtained for DL and centralized PM. Only the PM was published in the DSA, and individual pt information never left the originating institution. DSA interface allows clinicians to feed the requested pt characteristics into the DSA, and directly get the maculopathy free survival curve associated with the entered characteristics (figure 1).

Table 1			
	Covariate Name	Beta	p-value
Distributed Learning PM	Volume	3,6	0,004
	Diabetes	1,04	0,006
	Distance to Fovea	-0,18	9,8*10 ⁻⁶
Centralized PM	Volume	3,6	0,004
	Diabetes	1,04	0,006
	Distance to Fovea	-0,18	8,9*10 ⁻⁶

Conclusion

The DL Ecosystem, combined using the existing RLI and the developed DSA, enables Really Rapid Learning and application of PMs (PMID: 23993399) and allows training PMs by using geographically distributed DB without sharing clinical data. Future work pertains improving this RLI for ease of use, and iteratively re-learning PMs when new/more patient data and/or new sites enter the research. Implementing a preliminary analysis of new data is an essential step to study the consistency and to assess the degree of transferability of the updated PM.

Figure 1



EP-2255 High Dose Rate Brachytherapy In Patients With Non Melanoma Skin Cancer

P. Maia¹, A. Machado¹, M. Chen¹, M.L. Silva¹, R. Fogaroli¹, D. Castro¹, T. Coelho¹, H. Ramos¹, G. Gondim¹, A.C. Pellizzon¹

¹AC Camargo Cancer Center, Radiotherapy, sao paulo, Brazil

Purpose or Objective

Radiotherapy (RT) is a useful modality for the treatment of early stage non melanoma skin cancer (NMSC), in special where resection may produce a poor cosmetic result. RT for these tumors has been done, generally, with low-energy X-rays (120kV or less) or electrons of a linear accelerator (LINAC), because megavoltage photon RT has a skin-sparing property. Advances in technology and the commercial production of Leipzig applicators have allowed High Dose Rate after-load brachytherapy (HDR) to address a number of the challenges associated with the delivery of superficial radiation.

Material and Methods

This is a retrospective study of patients referred to the Radiation Oncology Department at AC Camargo Cancer Center with biopsy proven localized NMSC - basal cell carcinoma (BCC) or squamous cell carcinoma (SCC), with less than 40 mm in the biggest diameter and no more than 5 mm thickness. Patients who refused or unsuitable for surgery and those operated who presented adverse factors after local excision were also considered for the